# Shunsuke Kurita

PhD Student · Physics Program

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## Education

**Hiroshima University** *Hiroshima, Japan*

Ph.D. in Physics *Apr. 2025 - Present*

- Advisor: Prof. Kenta Shigaki
- Next-Generation Research Fellow

**Hiroshima University** *Hiroshima, Japan*

M.Sc. in Physics *Apr. 2023 - Mar. 2025*

- Advisor: Prof. Kenta Shigaki
- Thesis: "Azimuthal correlation analysis of electron-muon pairs for heavy-flavor measurements in proton-proton collisions at $\sqrt{s}$ = 13.6 TeV with ALICE"

**Hiroshima University** *Hiroshima, Japan*

B.Sc. in Physics *Apr. 2019 - Mar. 2023*

- Advisor: Prof. Kenta Shigaki
- Thesis: "Feasibility study of GeV-region muon identification by Ring-imaging Cherenkov detector using high spatial resolution silicon photomultiplier"

## Professional Experience

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| Aug. 2025 - Present | **ALICE Junior Representative for Japan**, ALICE Collaboration |
| Apr. 2023 - Present | **Member of the ALICE Collaboration**, CERN |
| Apr. 2024 - Present | **Researcher**, WPI-SKCM$^2$, Hiroshima University |

## Awards, Fellowships, & Grants

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| 2026 | **WPI-SKCM$^2$ Overseas Research Activity Support Program**, Hiroshima University |
| | **HU SPRING Research Study Abroad Program**, Hiroshima University |
| 2025 | **WPI-SKCM$^2$ Overseas Research Activity Support Program**, Hiroshima University |
| 2025 | **Student Presentation Award**, The Physical Society of Japan (JPS) |
| | **Next-Generation Research Fellowship**, Hiroshima University |

## Presentations

** presenting author*

### Contributed Presentations

**Kurita, S.*** (for the ALICE Collaboration). 2026. Transverse-momentum-differential azimuthal correlation of heavy-flavor electron-muon pairs in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE. Oral presentation: The 11th Asian Triangle Heavy-Ion Conference (ATHIC 2026), Shanghai, China. (Scheduled)

**Kurita, S.*** (for the ALICE Collaboration). 2026. Azimuthal correlation analysis of e-$\mu$ pairs from heavy-flavor decays in pp and OO collisions with ALICE. Oral presentation: The Physical Society of Japan (JPS) Autumn Meeting, Tokyo, Japan.

**Kurita, S.*** (for the ALICE Collaboration). 2026. Electron–muon azimuthal correlations from heavy-flavor hadron decays in proton-proton collisions at $\sqrt{s} = 13.6$ TeV with ALICE. Poster presentation: Hard Probes 2026, Nashville, TN, USA.

**Kurita, S.*** (for the ALICE Collaboration). 2025. Study of electron-muon correlations from heavy-flavor decays in proton-proton collisions with ALICE. Oral presentation: The Physical Society of Japan (JPS) Autumn Meeting, Hiroshima, Japan.

**Kurita, S.*** (for the ALICE Collaboration). 2024. Fundamental analysis for electron-muon azimuthal correlations from heavy-flavor decays in proton-proton collisions with ALICE. Oral presentation: The Physical Society of Japan (JPS) Annual Meeting, Hokkaido, Japan.

**Kurita, S.*** (for the ALICE Collaboration). 2024. First study for azimuthal correlations of electron-muon pairs from heavy flavor decays in proton-proton collisions with ALICE. Poster presentation: 12th International Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions (Hard Probes 2024), Nagasaki, Japan.

**Kurita, S.*** (for the ALICE 3 RICH Collaboration). 2023. Simulation study of the Ring Imaging Cherenkov detector for the next-generation heavy-ion experiment at LHC. Poster presentation: The 9th Asian Triangle Heavy-Ion Conference (ATHIC 2023), Hiroshima, Japan.

## Teaching Experience

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| Oct. 2023 - Mar. 2024 | **Numerical Methods in Physics**, Teaching Assistant |

## Technical Skills

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| Programming | **C/C++, Shell Scripting (Fish)**, |
| Physics Software | **ROOT, RooFit, ALICE O2Physics**, |
| Development Tools | **macOS, Linux environments, Git, Neovim, Zellij, iTerm2, Ghostty**, |
| Hardware | **High-performance custom PC building for data analysis**, |